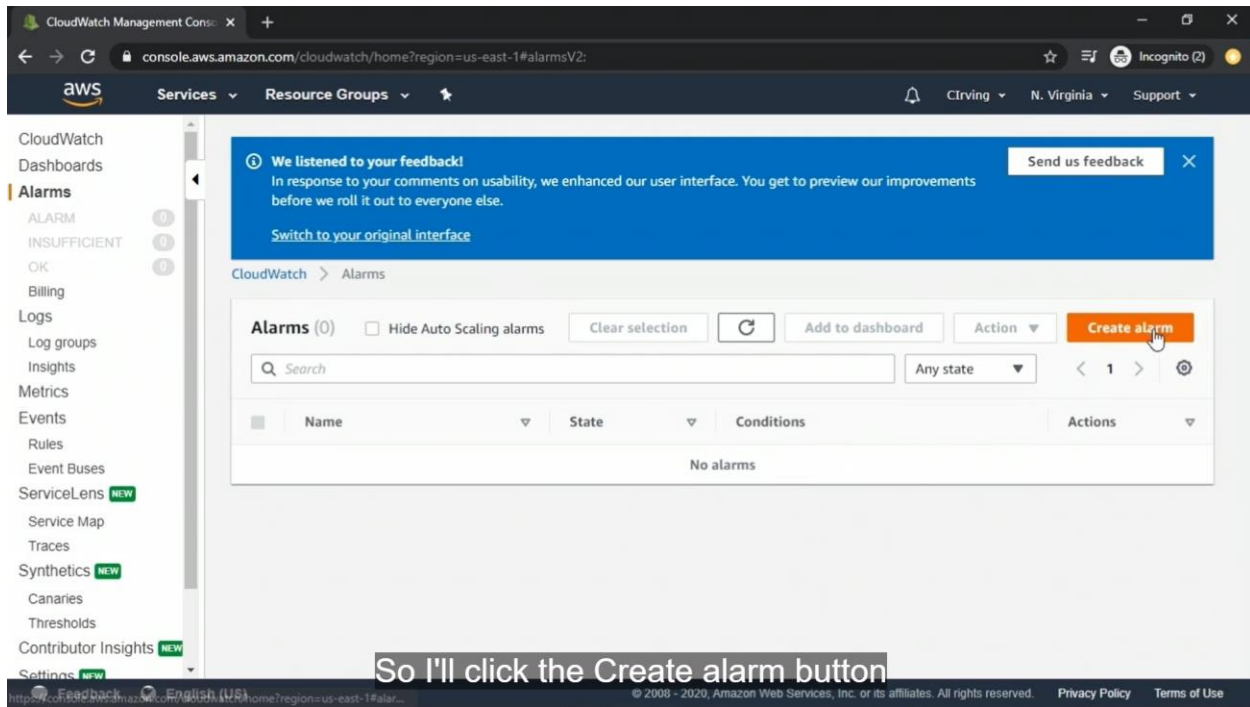
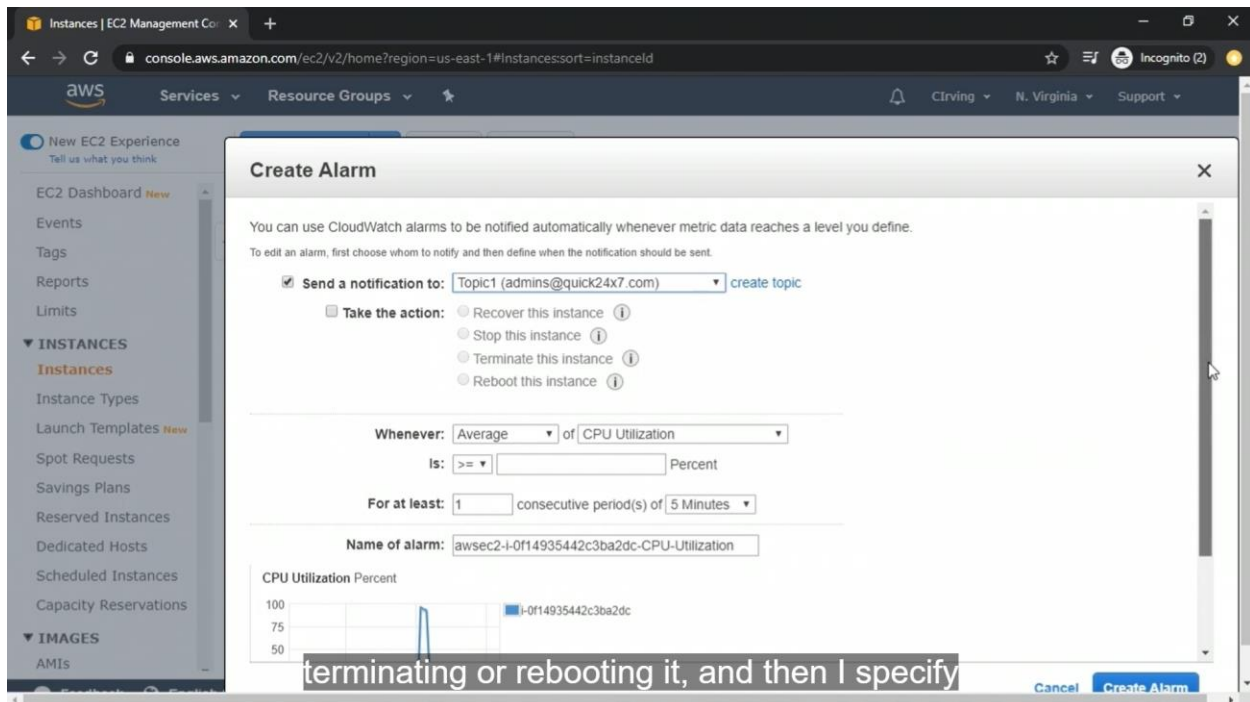


## 4.6 Resource Monitoring and Performance

### Configuring CloudWatch Alarms



### EC2 Instance Monitoring



## EC2 System Log

The screenshot shows the AWS Management Console with the EC2 System Log for instance `i-0a6ef83cd3a29cee2` (WinSrv2019-2). The log contains the following entries:

```
2020/02/11 10:50:21Z: Amazon SSM Agent v2.3.722.0 is running
2020/02/11 10:50:21Z: OsProductName: Windows Server 2019 Datacenter
2020/02/11 10:50:21Z: OsInstallOption: Full
2020/02/11 10:50:21Z: OsVersion: 10.0
2020/02/11 10:50:21Z: OsBuildLabEx: 17763.1.amd64fre.rs5_release.180914-1434
2020/02/11 10:50:24Z: Driver: AMS PV Network Device v8.2.5.32
2020/02/11 10:50:24Z: Drivers: AMS PV Storage Host Adapter v8.3.1.56
2020/02/11 10:51:29Z: Amazon SSM Agent v2.3.842.0 is running
2020/02/11 10:51:29Z: OsProductName: Windows Server 2019 Datacenter
2020/02/11 10:51:29Z: OsInstallOption: Full
2020/02/11 10:51:29Z: OsVersion: 10.0
2020/02/11 10:51:29Z: OsBuildLabEx: 17763.1.amd64fre.rs5_release.180914-1434
2020/02/11 10:51:35Z: Driver: AMS PV Network Device v8.2.5.32
2020/02/11 10:51:35Z: Drivers: AMS PV Storage Host Adapter v8.3.1.56
2020/02/11 10:54:03Z: Amazon SSM Agent v2.3.842.0 is running
2020/02/11 10:54:03Z: OsProductName: Windows Server 2019 Datacenter
2020/02/11 10:54:03Z: OsInstallOption: Full
2020/02/11 10:54:03Z: OsVersion: 10.0
2020/02/11 10:54:03Z: OsBuildLabEx: 17763.1.amd64fre.rs5_release.180914-1434
2020/02/11 10:54:09Z: Driver: AMS PV Network Device v8.2.5.32
2020/02/11 10:54:09Z: Drivers: AMS PV Storage Host Adapter v8.3.1.56
```

Below the log, a text overlay reads "about that specific, EC2 instance." with a blue "Close" button. At the bottom, a Windows Command Prompt shows the command:

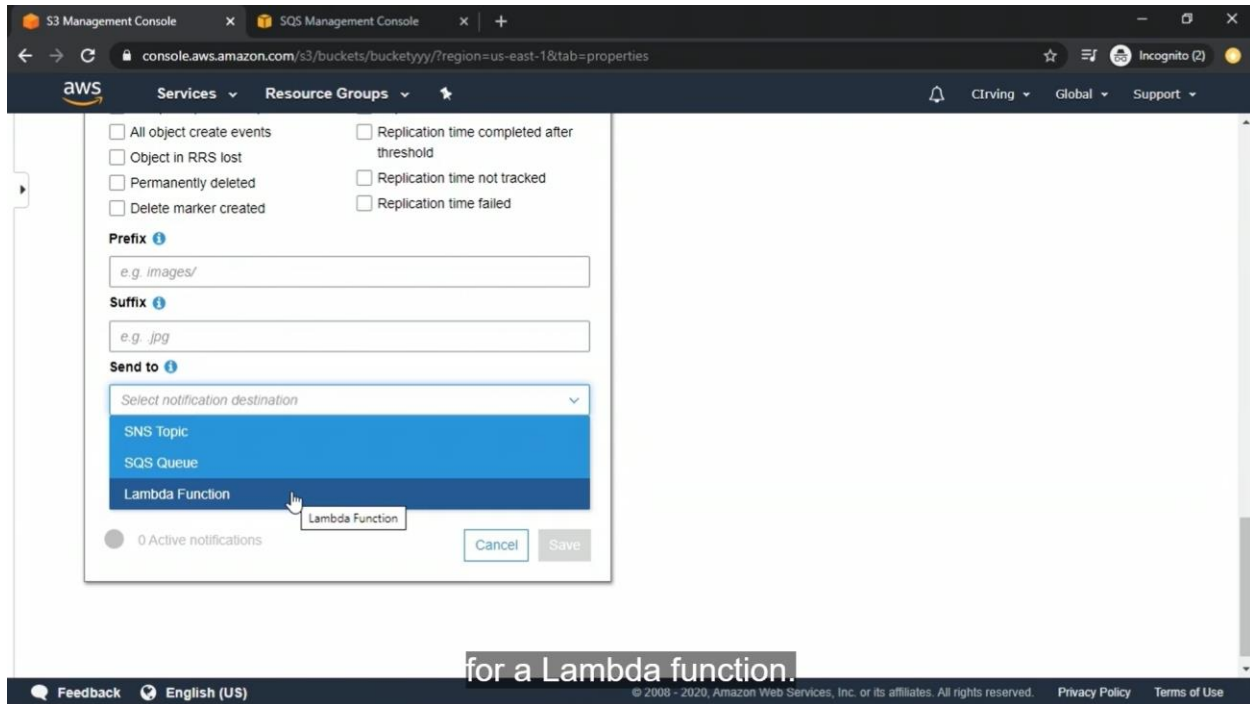
```
C:\>aws ec2 --output=text get-console-output --instance-id i-0a6ef83cd3a29cee2
```

## S3 Bucket Monitoring

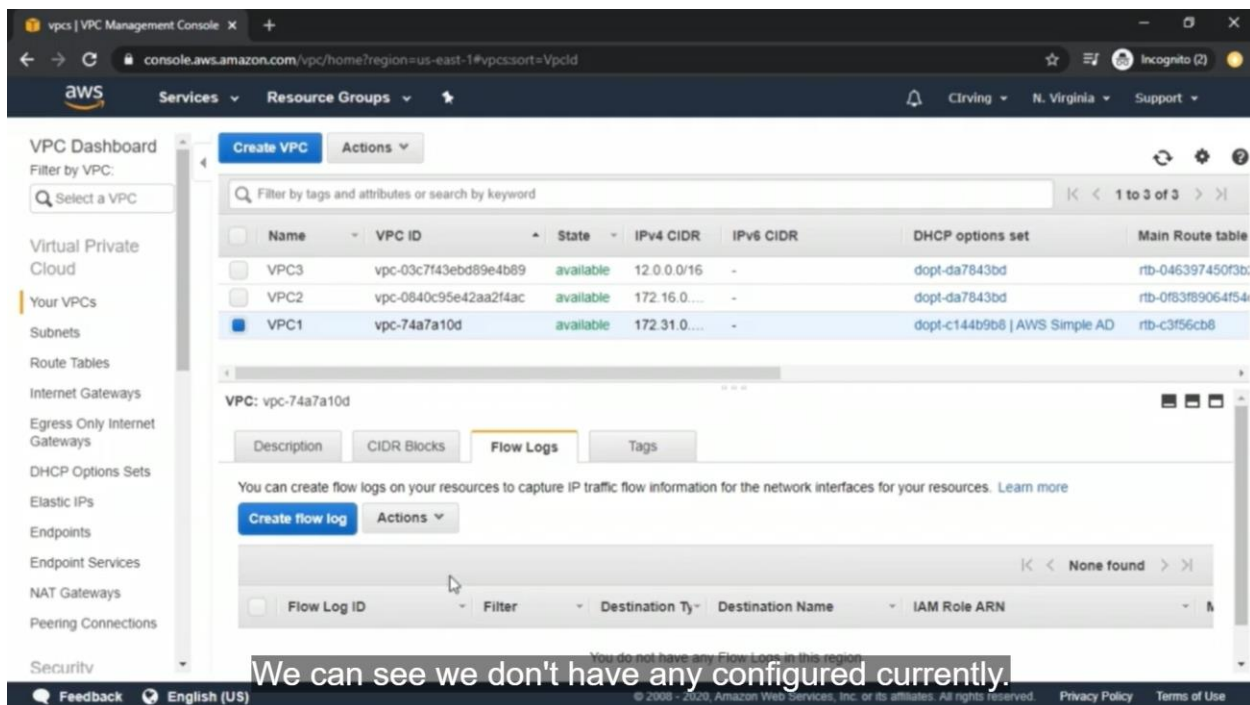
## S3 Bucket Storage Inventory

## S3 Server Access Logging

## S3 Bucket Event Monitoring

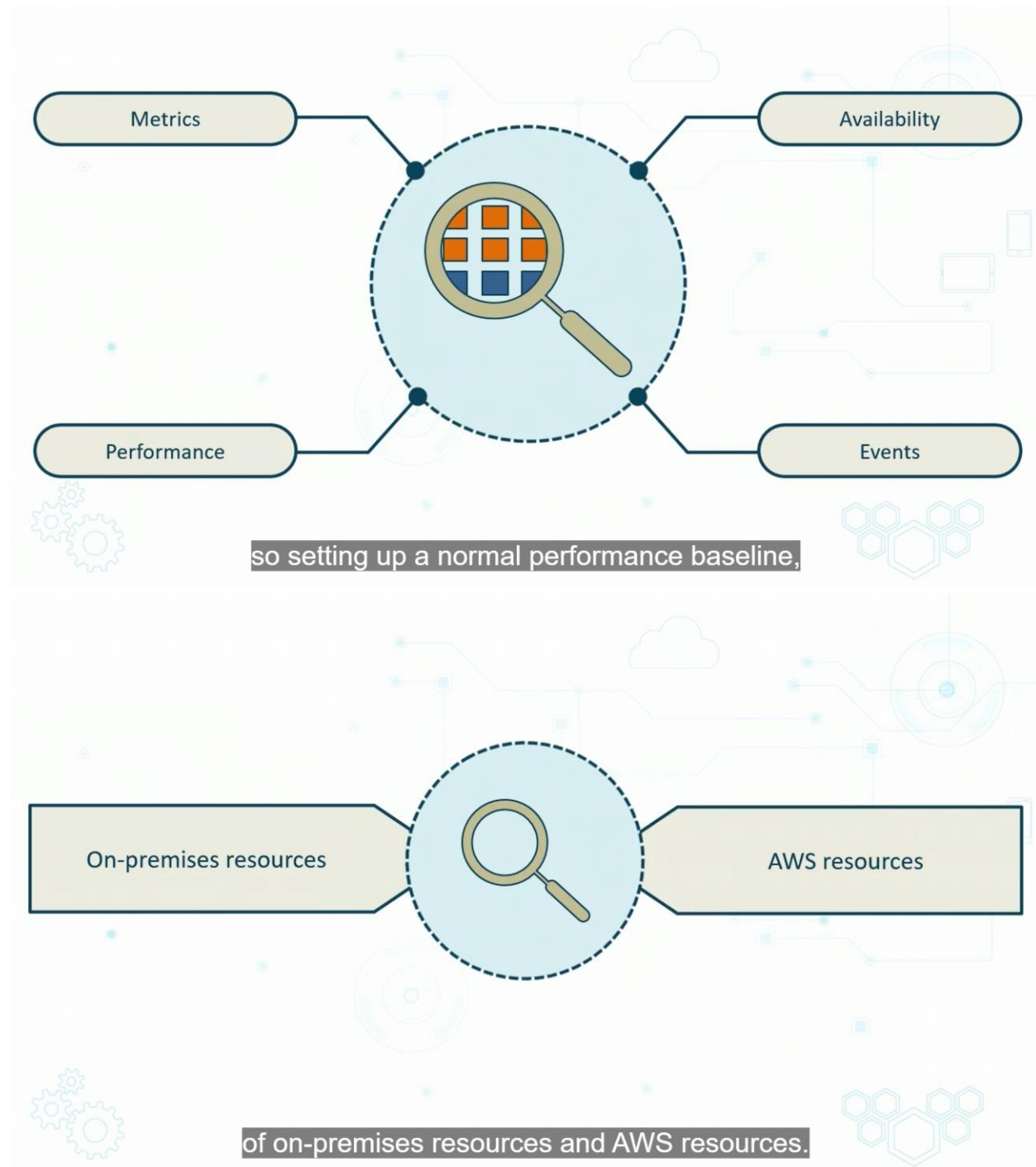


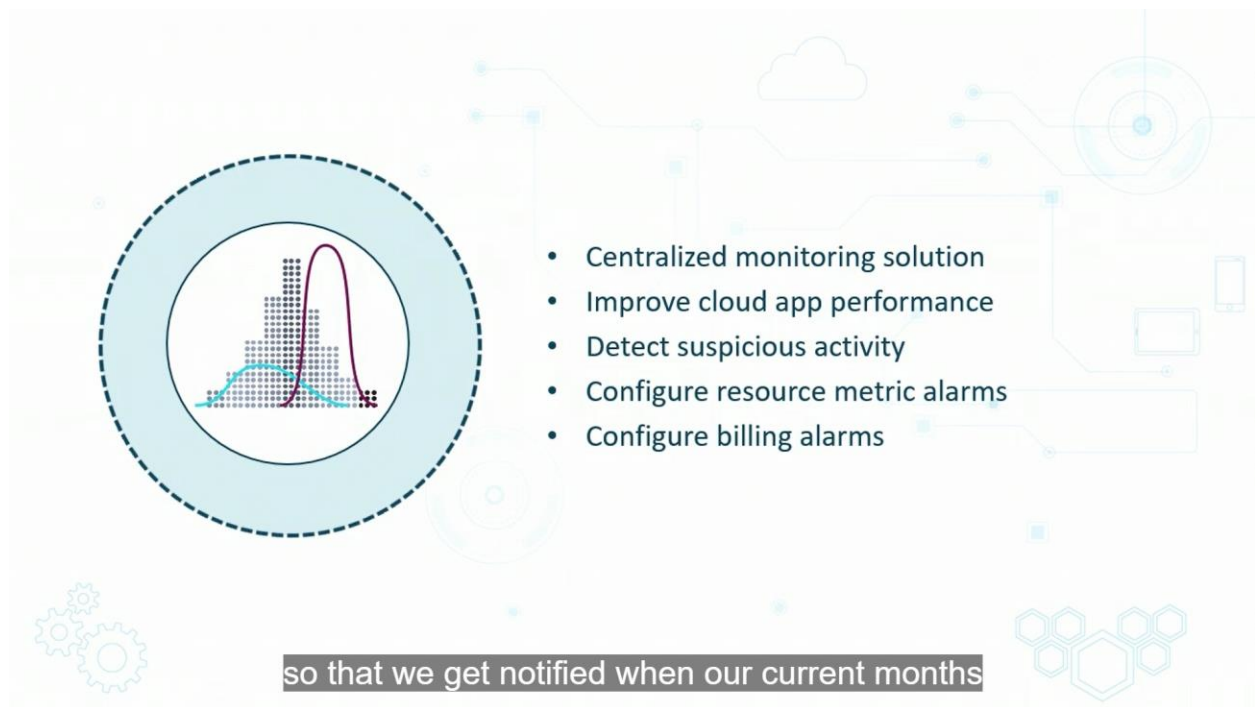
## VPC Flow Logs



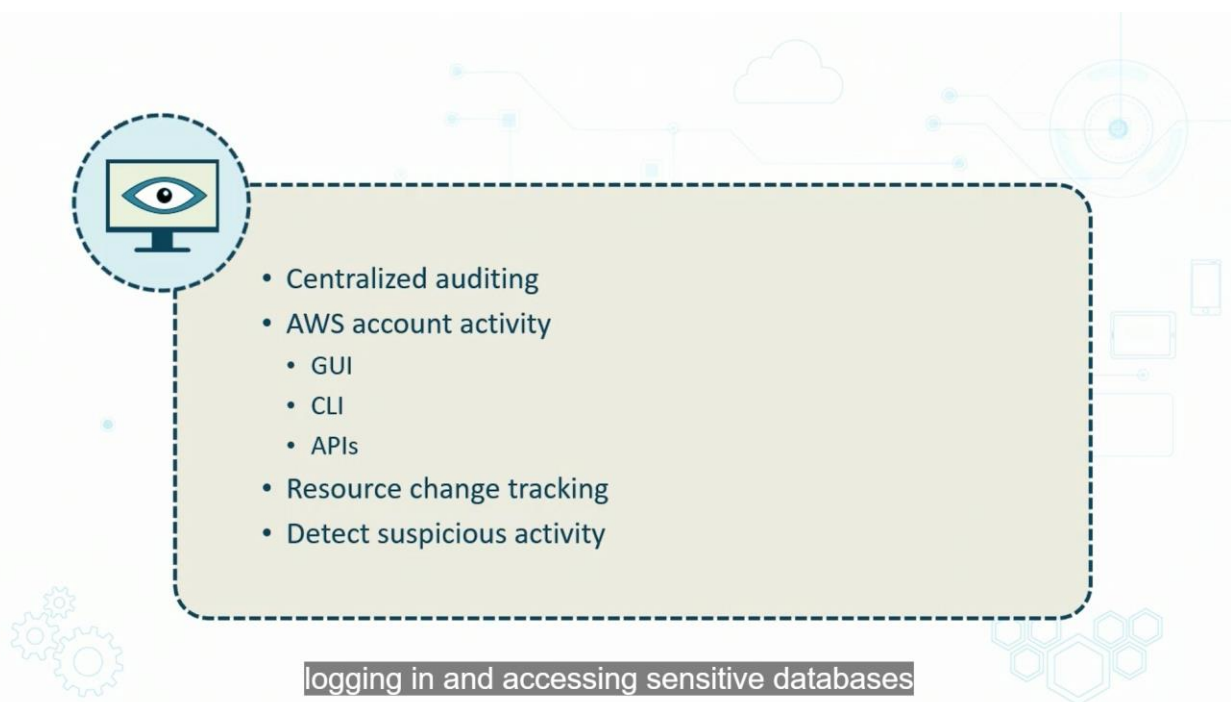
## AWS Monitoring Overview

Monitoring: CloudWatch





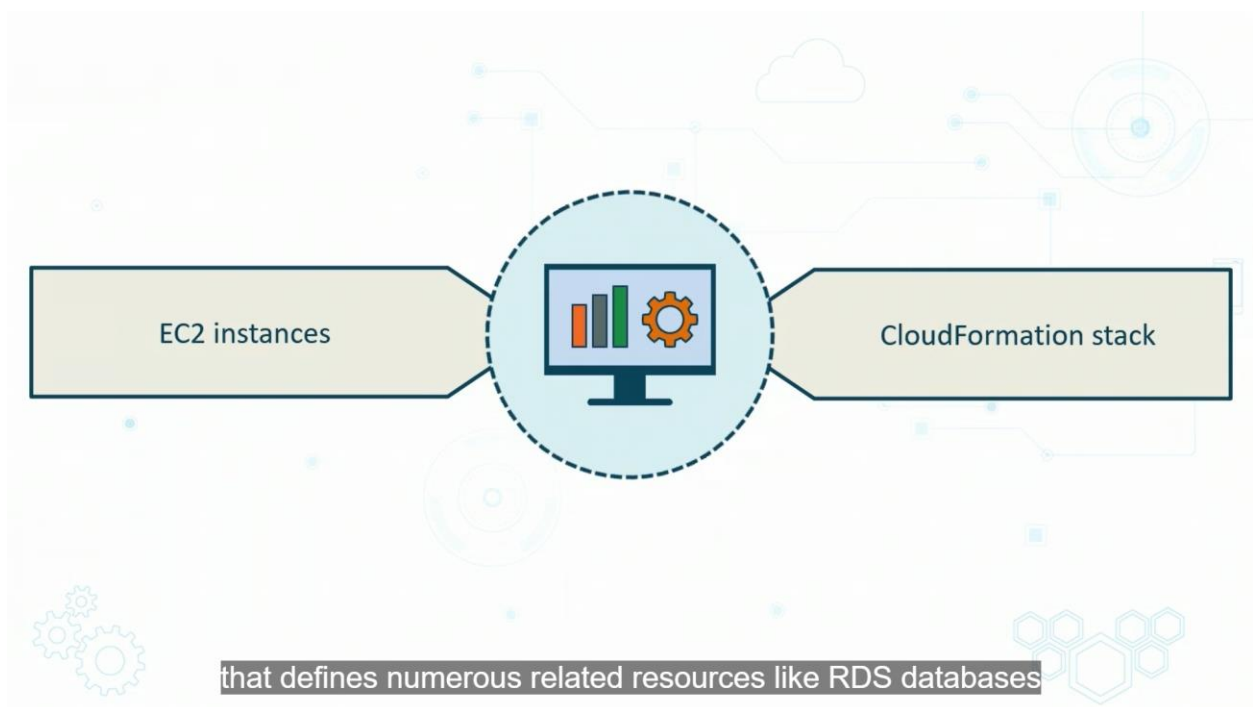
## Monitoring: CloudTrail

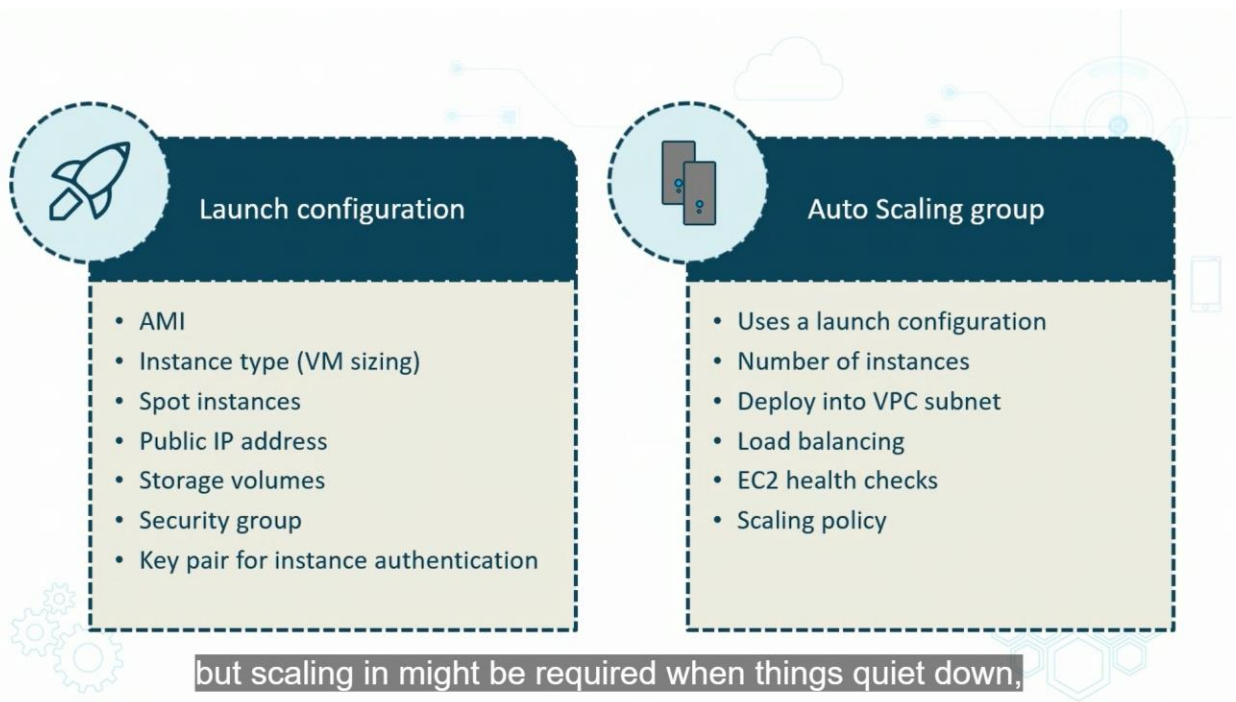




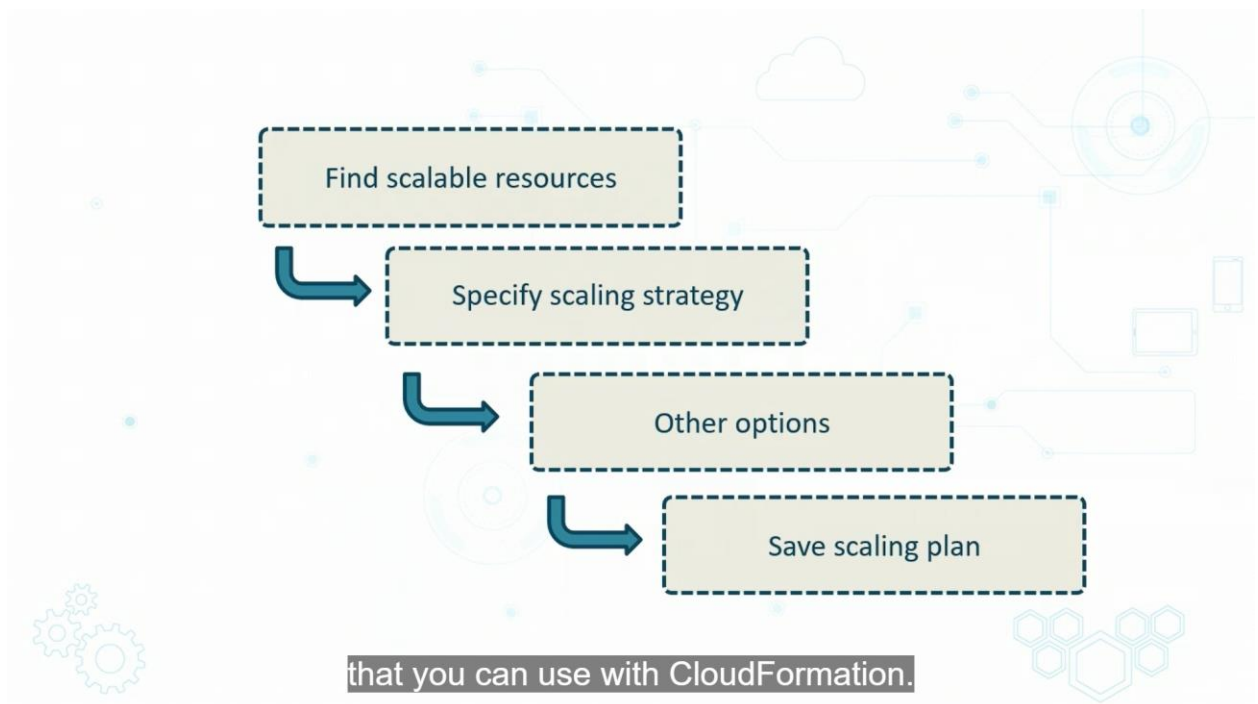
## Viewing CloudTrail Events

### Auto Scaling



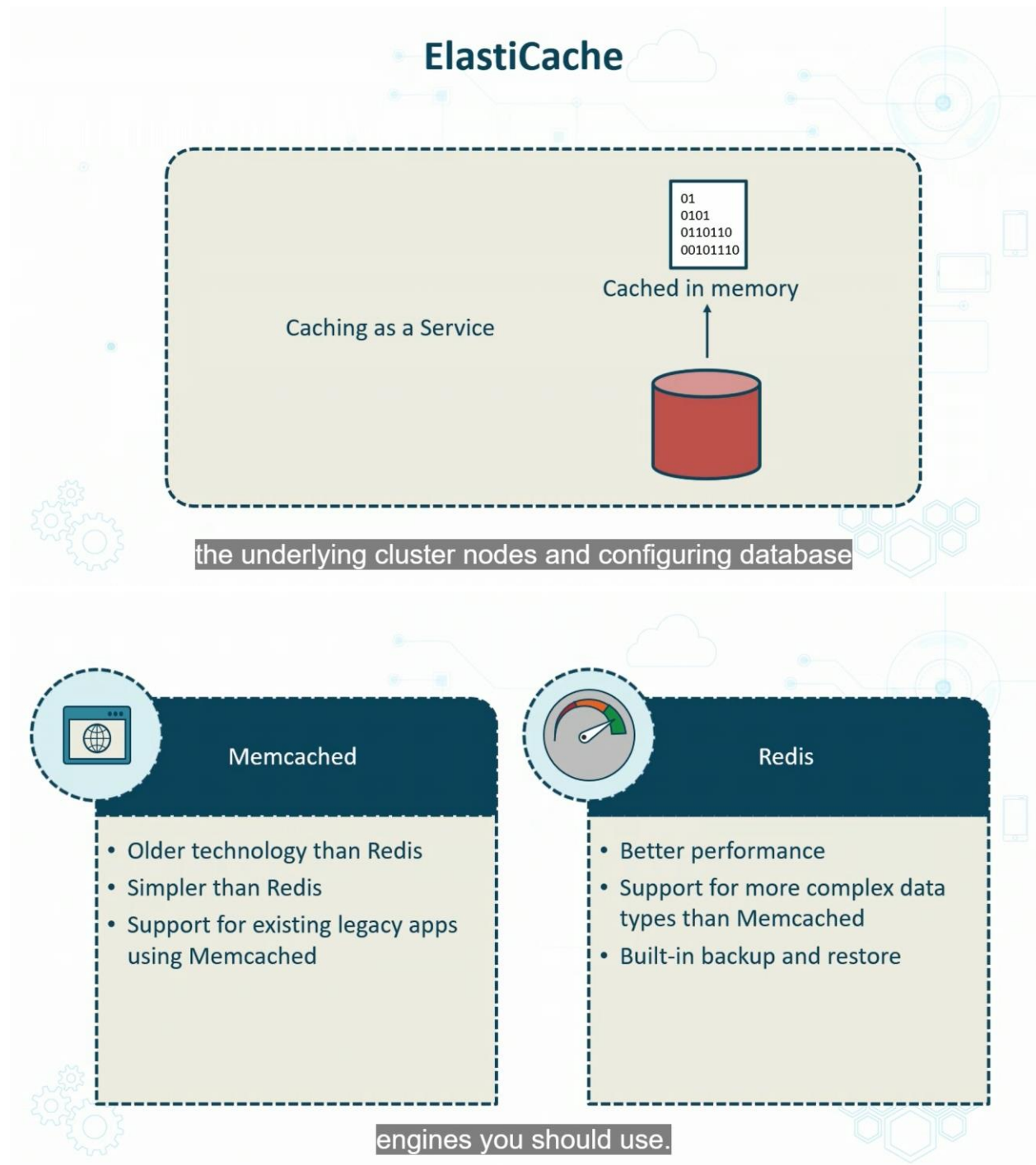


## CloudFormation Scaling Plans



## Enabling Auto Scaling

## ElastiCache

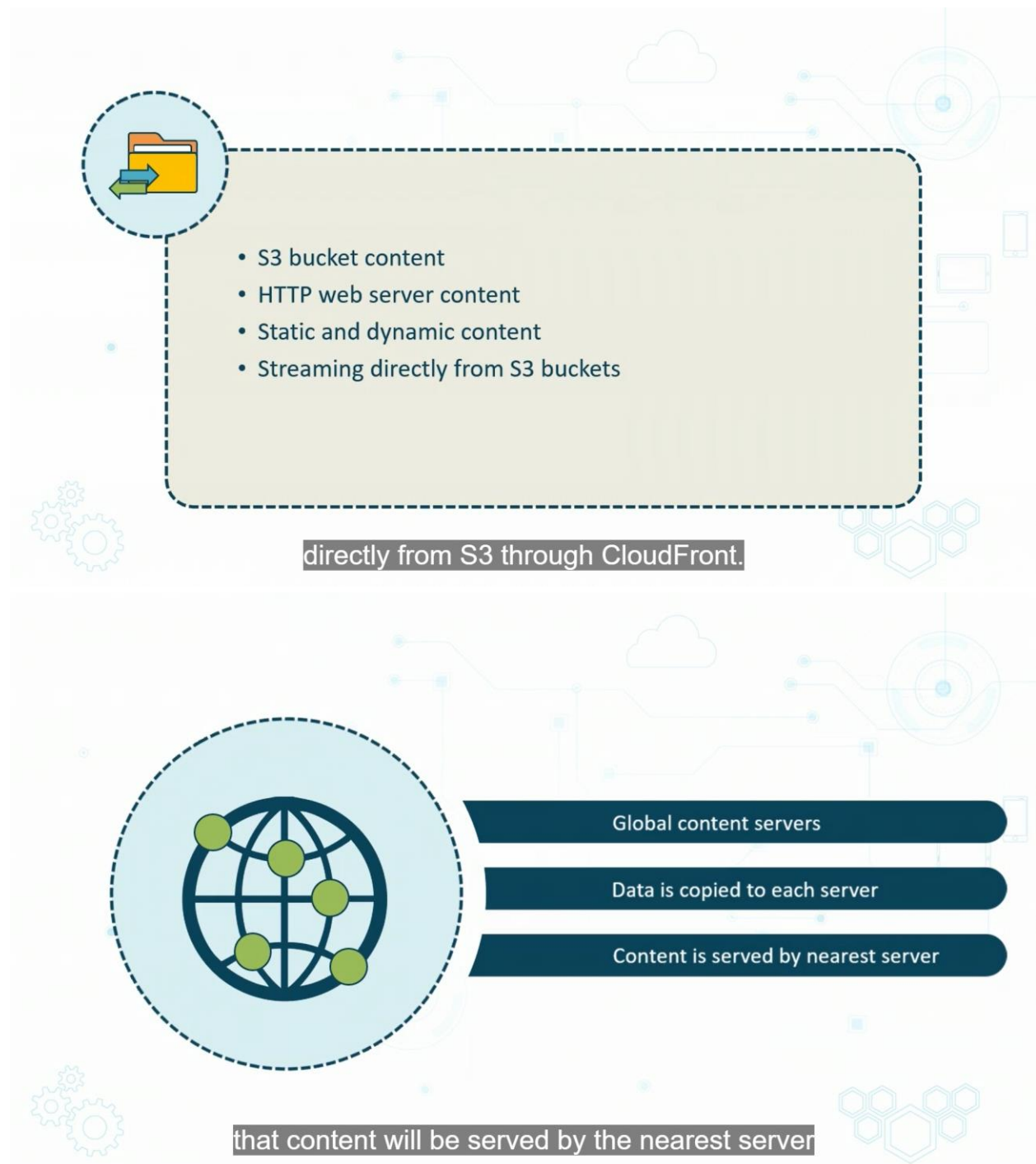


## ElastiCache Deployment with Memcached

## ElastiCache Deployment with Redis

## CloudFront

### Content Delivery Networks (CDNs)





Suppose you are hosting a web app in Western US

- A user in Ireland accesses your app
- Rather than additional latency for each file requested (e.g. 100 ms), the request is routed to the Dublin edge location (e.g. <10 ms)

for each file that gets requested by that user,

## CloudFront Configuration

AWS CloudFront Management Console - console.aws.amazon.com/cloudfront/home#create-distribution:

Step 1: Select delivery method  
Step 2: Create distribution

CloudFront serves your content over HTTPS to clients that support SNI. SNI is supported by browsers and clients released after 2015. Additional charge for this option. [Learn More](#)

☐ Legacy Clients Support (\$600/month prorated charge applies. [Learn about pricing](#))  
CloudFront allocates dedicated IP addresses at each CloudFront edge location to serve your content over HTTPS. [Learn More](#)

**Security Policy**

- ☐ TLSv1
- ☐ TLSv1\_2016
- ☒ TLSv1.1\_2016 (recommended)
- ☐ TLSv1.2\_2018

[See the list of protocols and ciphers that CloudFront uses for each security policy.](#)

**Supported HTTP Versions**

- ☒ HTTP/2, HTTP/1.1, HTTP/1.0
- ☐ HTTP/1.1, HTTP/1.0

**Default Root Object**

**Logging**

- ☐ On
- ☒ Off

**Bucket for Logs**

**Log Prefix**

a lot of web browser makers are stopping support for anything

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1. Why would you select a Memcached ElastiCache cluster over Redis?
  - Legacy app support
  - Simple configuration
  
2. Which are valid CloudWatch alarm states?
  - OK
  - In alarm
  - Insufficient data
  
3. Which ElastiCache option provides the simplest configuration options?
  - Redis
  
4. Which statement regarding S3 bucket server access logging is correct?
  - Logging is disabled by default
  
5. Which is a valid option for S3 inventory report file formats?
  - CSV
  
6. Which types of CloudFront CDNs can be deployed?
  - Web
  - RTMP
  
7. What can be done when you need to capture VPC network traffic? The solution must involve the least amount of administrative effort possible.
  - Enable a flow log
  
8. You are in the EC2 instance view. Where should you click to retrieve an instance system log?
  - Actions

9. Which word is the most closely related to CloudTrail?
- Auditing
10. You need to ensure that CloudTrail results are sent to an S3 bucket. What should you configure?
- Trail
11. You would like to configure S3 bucket event notification. Using the console, you have clicked on the S3 bucket name. What should you click next?
- Properties
12. What is the benefit of a content delivery network (CDN)?
- Reduced network latency
13. What is the purpose of an EC2 launch configuration?
- EC2 instance details for auto scaling
14. You are viewing S3 bucket metrics through the S3 management console. Which metrics are shown by default?
- BucketSizeBytes
  - NumberofObjects
15. You would like to receive EC2 instance metric data every minute. What should you do?
- Enable detailed monitoring
16. Why would you select a Memcached Redis cluster over Memcache?
- Better performance than Redis
  - Built-in backup and restore
17. What type of scaling is AWS EC2 auto scaling?
- Horizontal
  - Scaling out